

Module 10: FIFO Page Replacement and Belady's Anomaly

AKTU BCS401

Unit 4

Page Replacement Paradox

1. FIFO Algorithm & Dirty Bit

FIFO memory me sabse pehle enter karne wale page ko replace karta hai. Dirty Bit (0/1) indicate karta hai ki page modify hua hai ya nahi; clean page evict karte samay disk write save ho jata hai.

2. Belady's Anomaly Proof

Normally frames badhane se faults kam hote hain, lekin FIFO me frames 3 se 4 karne par faults 9 se badhkar 10 ho jate hain!

Belady's Anomaly: FIFO 3 Frames vs 4 Frames Proof

Reference String: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

FIFO with 3 Frames (9 Page Faults)

Initial: [Empty]

Refs 1, 2, 3: Frames = [1, 2, 3] → 3 Faults

Ref 4: Replace 1 → [4, 2, 3] → Fault 4

Ref 1: Replace 2 → [4, 1, 3] → Fault 5

Ref 2: Replace 3 → [4, 1, 2] → Fault 6

Ref 5: Replace 4 → [5, 1, 2] → Fault 7

Ref 1: HIT! Frames = [5, 1, 2]

Ref 2: HIT! Frames = [5, 1, 2]

Ref 3: Replace 1 → [5, 3, 2] → Fault 8

Ref 4: Replace 2 → [5, 3, 4] → Fault 9

Ref 5: HIT! Frames = [5, 3, 4]

Total Page Faults = 9

FIFO with 4 Frames (10 Page Faults!)

Refs 1, 2, 3, 4: Frames = [1, 2, 3, 4] → 4 Faults

Ref 1: HIT! Frames = [1, 2, 3, 4]

Ref 2: HIT! Frames = [1, 2, 3, 4]

Ref 5: Replace 1 → [5, 2, 3, 4] → Fault 5

Ref 1: Replace 2 → [5, 1, 3, 4] → Fault 6

Ref 2: Replace 3 → [5, 1, 2, 4] → Fault 7

Ref 3: Replace 4 → [5, 1, 2, 3] → Fault 8

Ref 4: Replace 5 → [4, 1, 2, 3] → Fault 9

Ref 5: Replace 1 → [4, 5, 2, 3] → Fault 10

Total Page Faults = 10 (Faults BADH GAYE!)

Frames 3 → 4 badhane par Faults 9 → 10 ho gaye!
• FIFO is NOT a Stack Algorithm.

3. Summary Comparison

Why it happens: FIFO is NOT a Stack Algorithm ($S_n \not\subseteq S_{n+1}$).

Stack Algorithms: LRU and Optimal algorithms strictly follow subset property and NEVER suffer from Belady's Anomaly!